INCH-POUND

MIL-DTL-26521K
INTERIM AMENDMENT 1 (CC)
18 July, 2002

DETAIL SPECIFICATION

HOSE ASSEMBLY, NONMETALLIC, FUEL, COLLAPSIBLE, LOW TEMPERATURE
WITH NON-REUSABLE COUPLINGS

This interim amendment is approved for use within the Defense Supply Center Columbus, Department of Defense, with MIL-DTL-26521K, dated 15 February 1999.

PAGE 8

3.5.5: Delete and substitute:

"3.5.5 Low temperature flexibility. The hose shall be flexible at temperatures ranging down to -40° ± 2° Fahrenheit (F). The manufacturer shall submit hose testing results indicating the maximum registered torque required to bend the hose around the test drum at a temperature of -40° ±2° F. Test results shall be reported in foot-pounds force (lb-ft). The test shall be conducted in accordance with 4.6.6.

PAGE 13

4.6.6: Delete and substitute:

"4.6.6 Low temperature flexibility. Test samples shall be pre-conditioned at -40° F for 24 hours prior to test. The flexibility tests shall be carried out with an empty length of hose, the length of which is calculated as follows:

Hose length = (3.142 X D) + (4 X d).

Where D = Test drum diameter
d = Hose diameter

Determine the external test drum diameter based on the hose diameter using table V."

Add the following table and figure after 4.6.6.

TABLE V. Test drum external diameter in relation to hose diameter.

Hose diameter (d) Inches	Test drum external diameter (D) Inches
2	20
2.5	25
3	30
4	40

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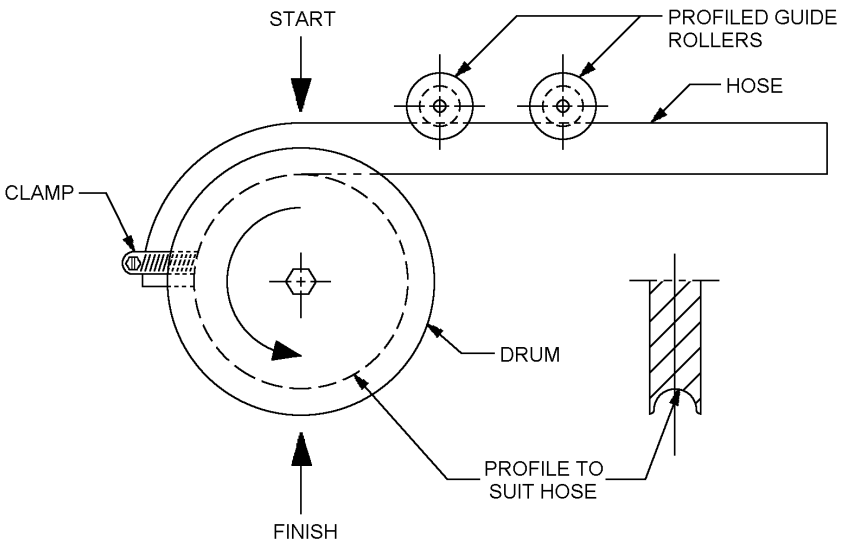

NOTES:

1. Determine the hose length and the external test drum diameter from table V and the equation (see 4.6.6).
2. Attach the hose to the test drum with the clamp as shown on figure 3.
3. Place the test drum and hose specimen in a cold box/bath to lower their temperature to -40° F. The test drum and specimen for hoses with inside diameters of 3 inches and smaller shall be kept in the cold box/bath during the test to ensure the test temperature is maintained. For the hose with inside diameter of 4 inches, the test drum and specimen must be at the test temperature but may be removed from the cold box/bath for the test procedure. In this case the test must be initiated within 30 seconds of removal from the cold box/bath.
4. Rotate the test drum using a torque wrench with a dial indicator. The drum must be rotated 180° within 10 seconds.
5. Record the maximum registered torque required to bend the hose around the test drum. Report the test result in lb-ft. Verify that the hose is flexible.
6. Proof test hose after this test. Then split hose lengthwise and inspect for cracks, breaks and layer separation. Failure of this test is cause for rejection of the production lot.

FIGURE 3. Test fixture for cold flexibility evaluation.

CONCLUDING MATERIAL

Preparing activity:
DLA-CC

(Project 4720-0344-001)

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